Introduction to Artificial Intelligence

Week 6

Traveling Salesman Problem (TSP)

- "Given a list of cities and the distances between each pair of cities, what is the shortest possible route that visits each city exactly once and returns to the origin city?"

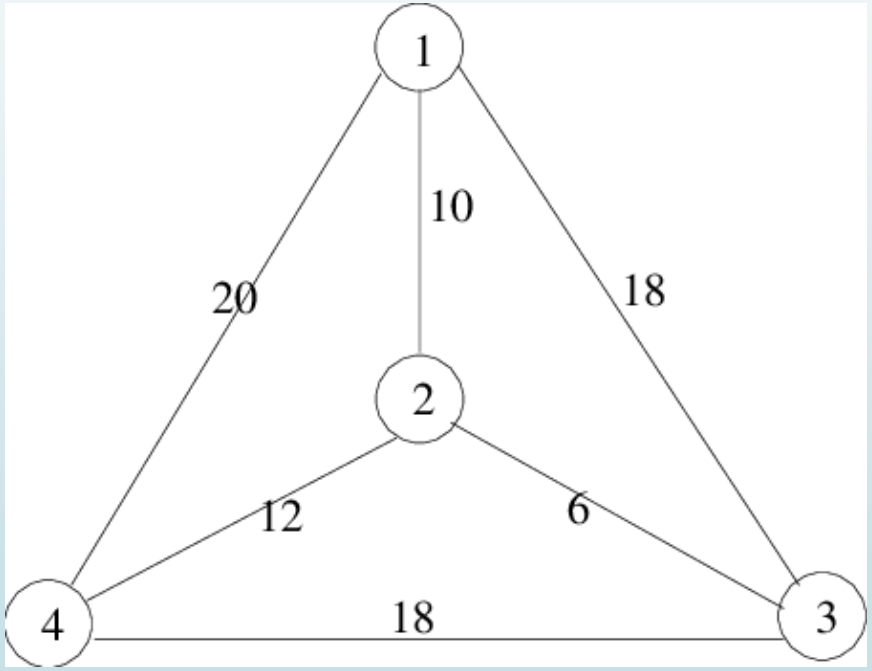


Traveling Salesman Problem (2)

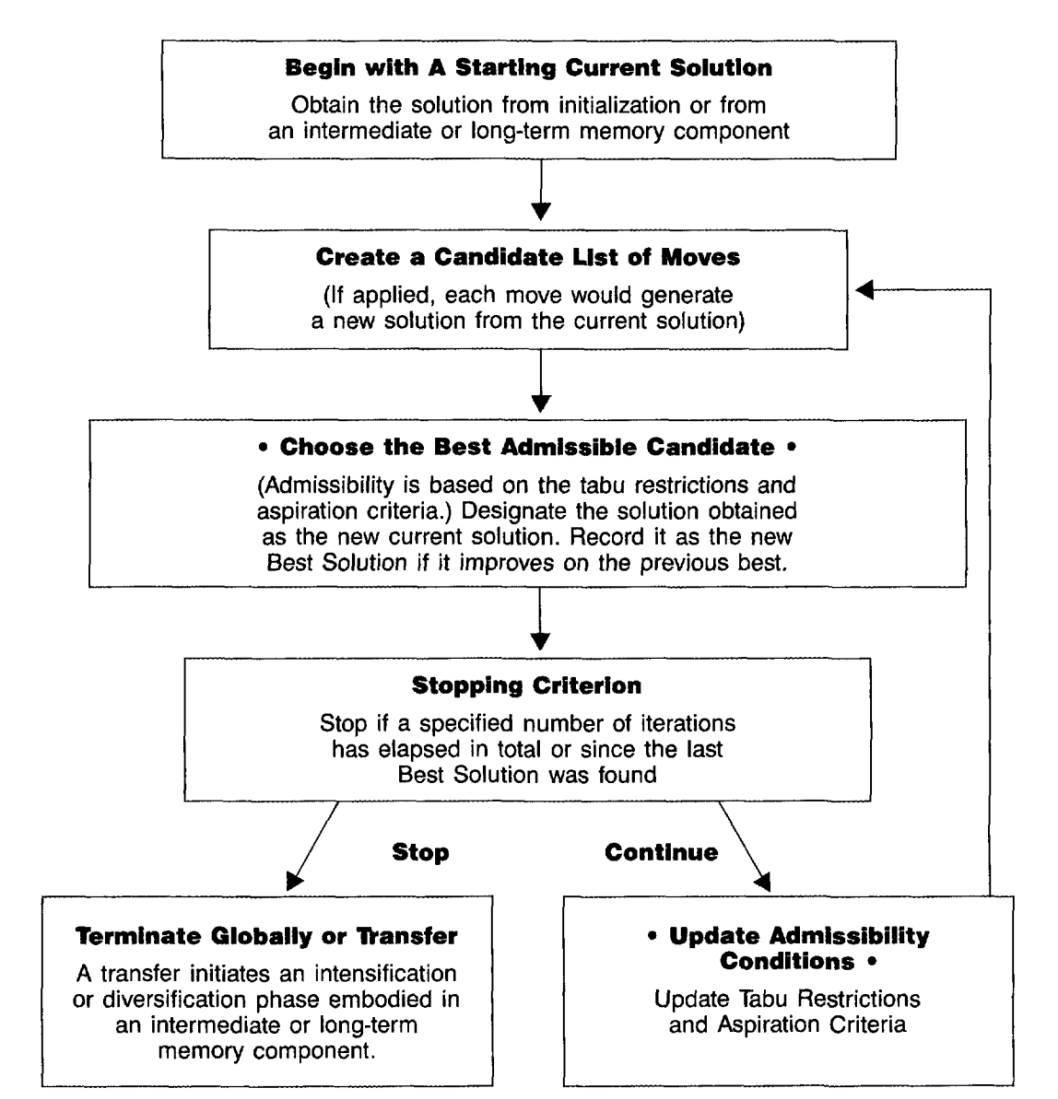
	C1	C2	C3	C4	C5
C1	0	132	217	164	158
C2	132	0	290	201	79
C3	217	290	0	113	303
C4	164	201	113	0	196
C5	158	79	303	196	0

The solution can be written as C2->C5->C1->C4->C3->C2. An itinerary that begins and ends at the same city and visits each city once is called a tour.

TSP – Graph Representation



Tabu Search


Begin with A Starting Current Solution
Obtain the solution from initialization or from an intermediate or long-term memory component
Create a Candidate List of Moves
(If applied, each move would generate a new solution from the current solution)
• Choose the Best Admissible Candidate •
(Admissibility is based on the tabu restrictions and aspiration criteria.) Designate the solution obtained as the new current solution. Record it as the new Best Solution if it improves on the previous best.
Stopping Criterion
Stop if a specified number of iterations has elapsed in total or since the last Best Solution was found
Stop
Continue
Terminate Globally or Transfer
A transfer initiates an intensification or diversification phase embodied in an intermediate or long-term memory component.
• Update Admissibility Conditions •
Update Tabu Restrictions and Aspiration Criteria

TSP – Tabu Search

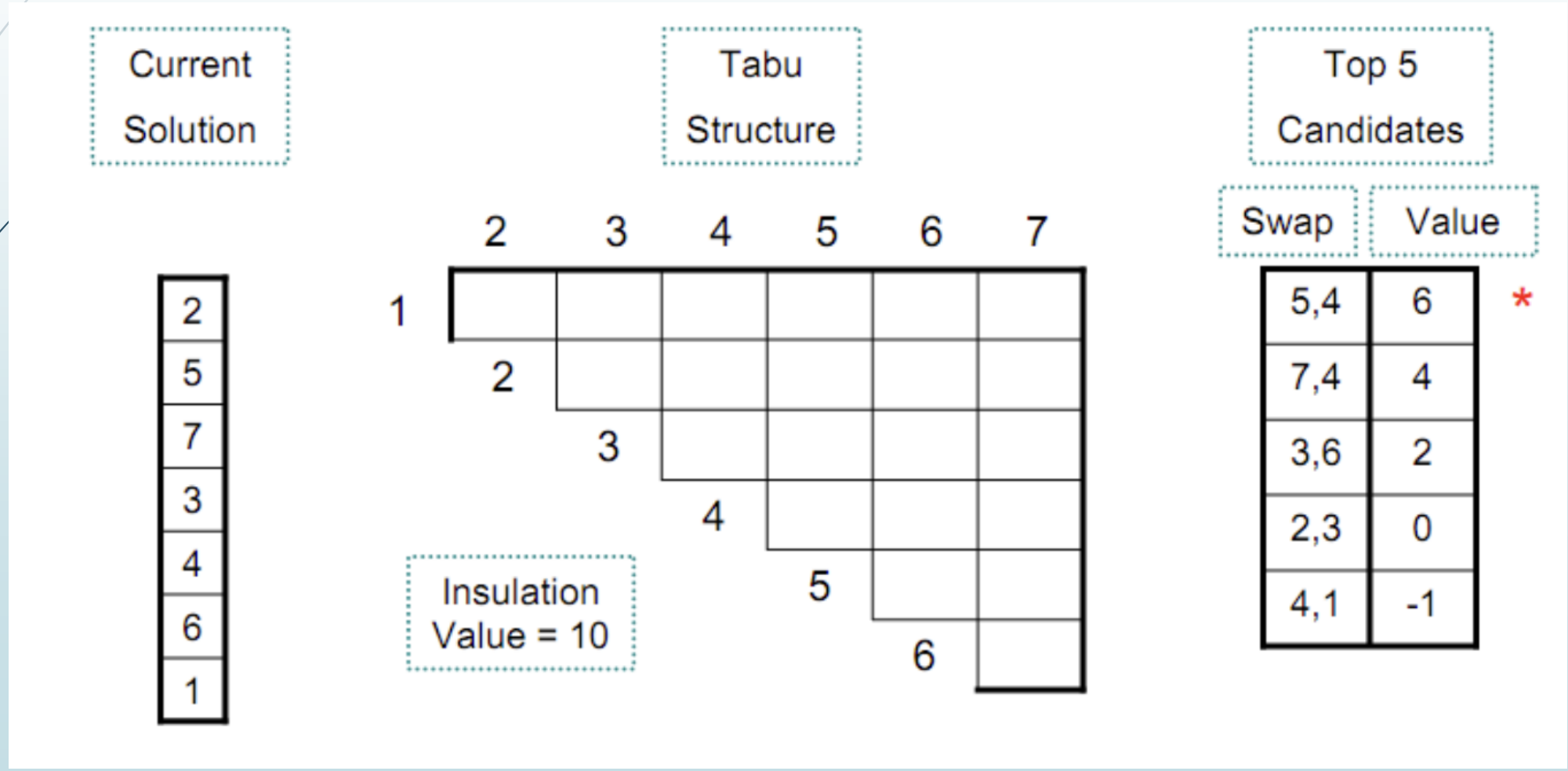
Suppose the initial solution to our problem is the shown as:

2	5	1	4	3	2

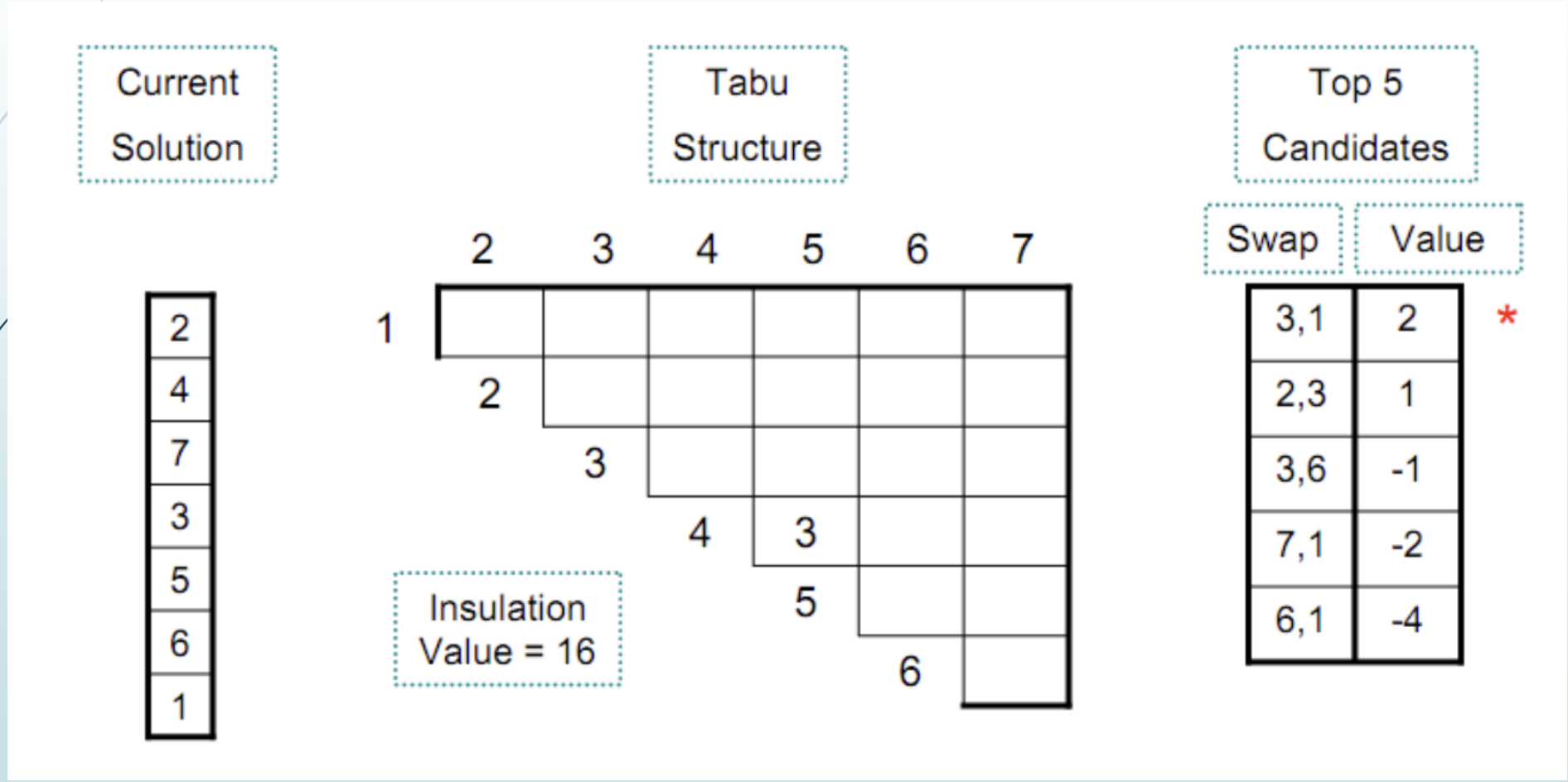
The ordering specifies that city 2 is placed in the first position, followed by city 5, etc. Tabu search methods operate to identify the neighbourhood that can be reached from any current solution. Identifying moves that lead from one solution to the next. In our problem, a swap exchanges the position of two cities.

2	4	1	5	3	2

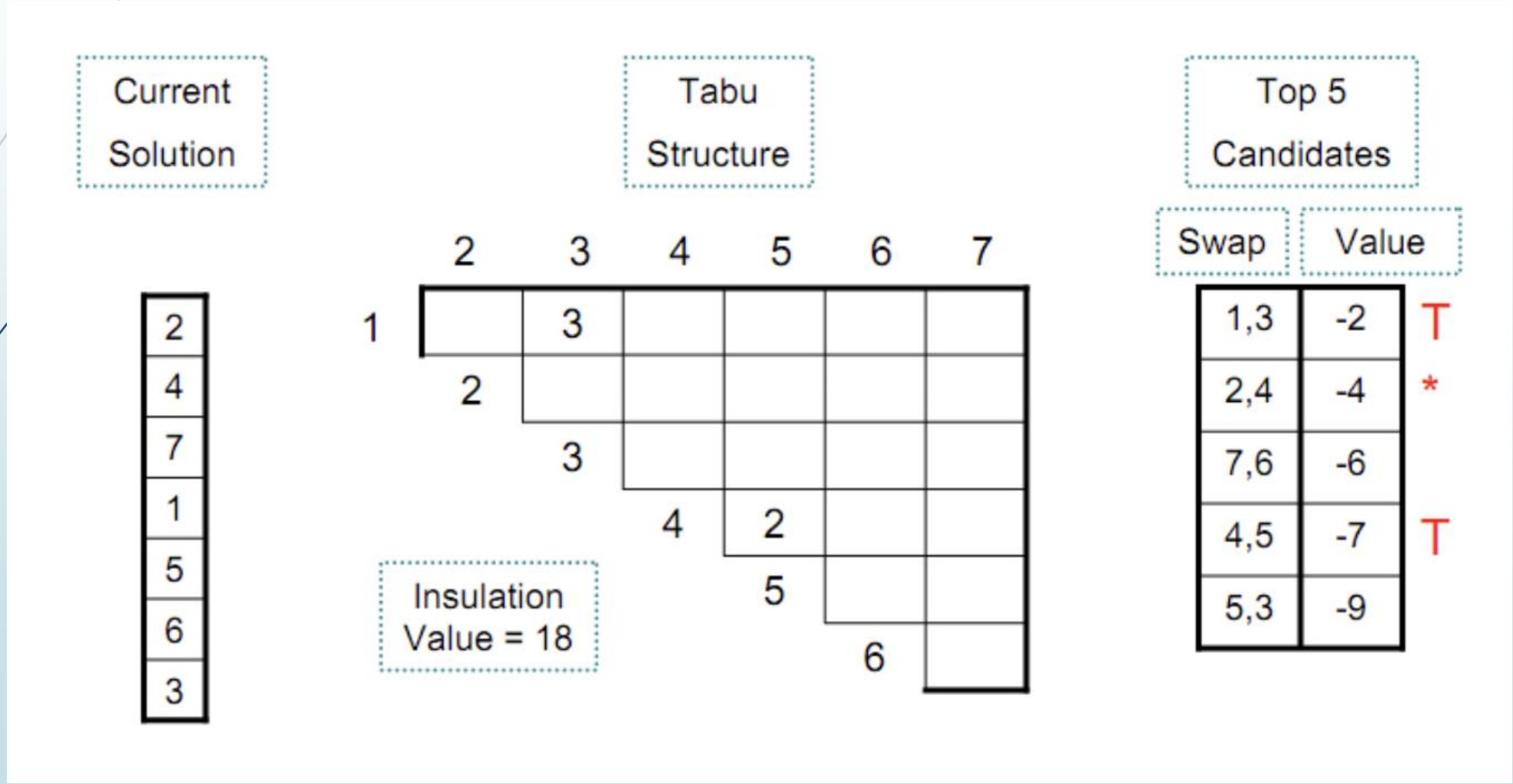
Resolution for the TSP using Tabu Search



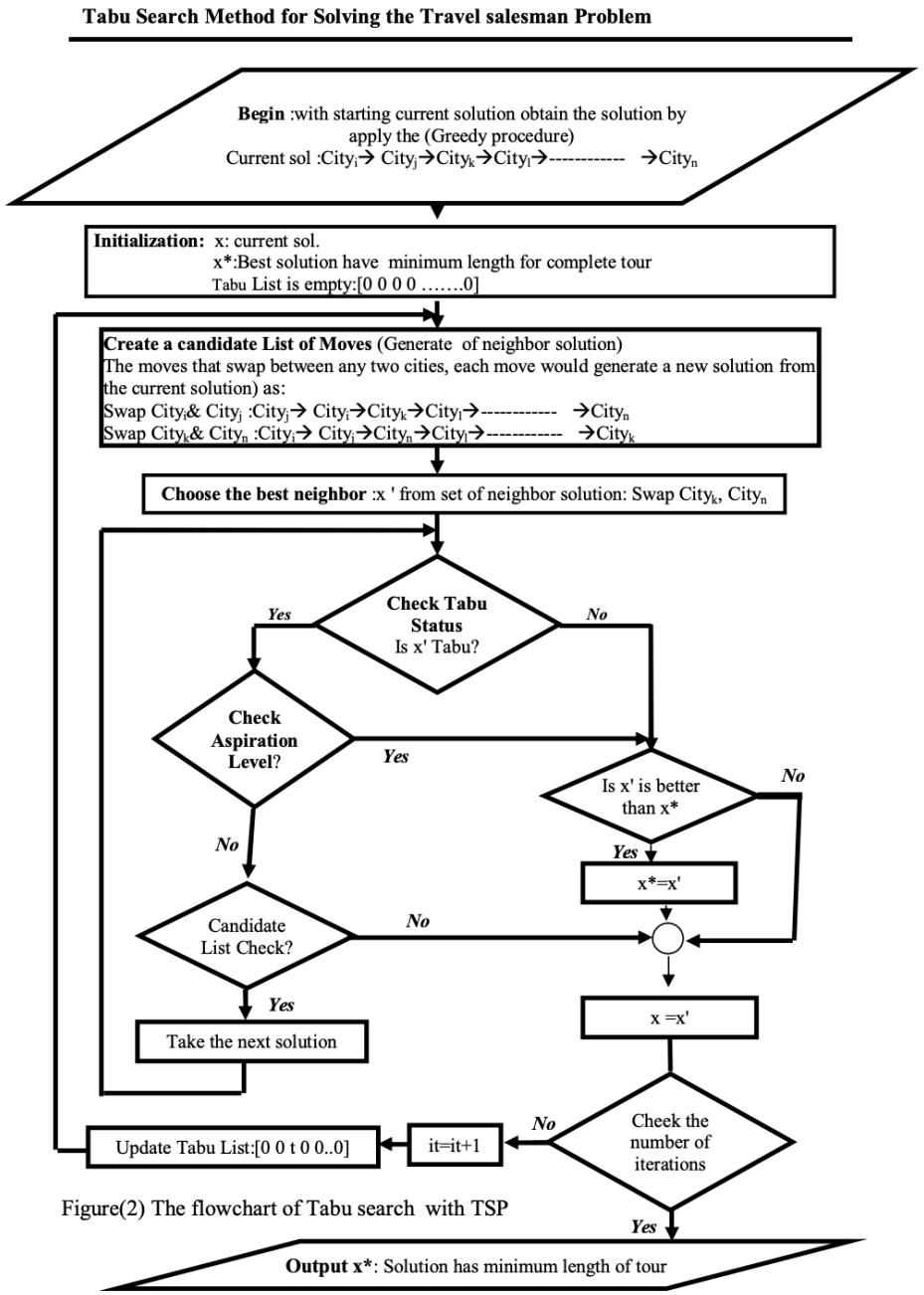
Resolution for the TSP using Tabu Search (2)



Resolution for the TSP using Tabu Search (3)



SOLUTION



Figure(2) The flowchart of Tabu search with TSP

Tabu Search Characteristics

Aspiration Criterion

In order to override the tabu list when there is a good tabu move, aspiration criterion is used, the tabu move is accepted if it produces better solution than the best obtained so far.

Termination criterion

The most commonly used stopping criteria in tabu search are:

1- Terminate the search after number of iterations.

2- After some number of iterations without an improvement in the objective function value.

3- When the objective reaches a pre-specified threshold value. In this research the first criteria is used for stopping criteria.

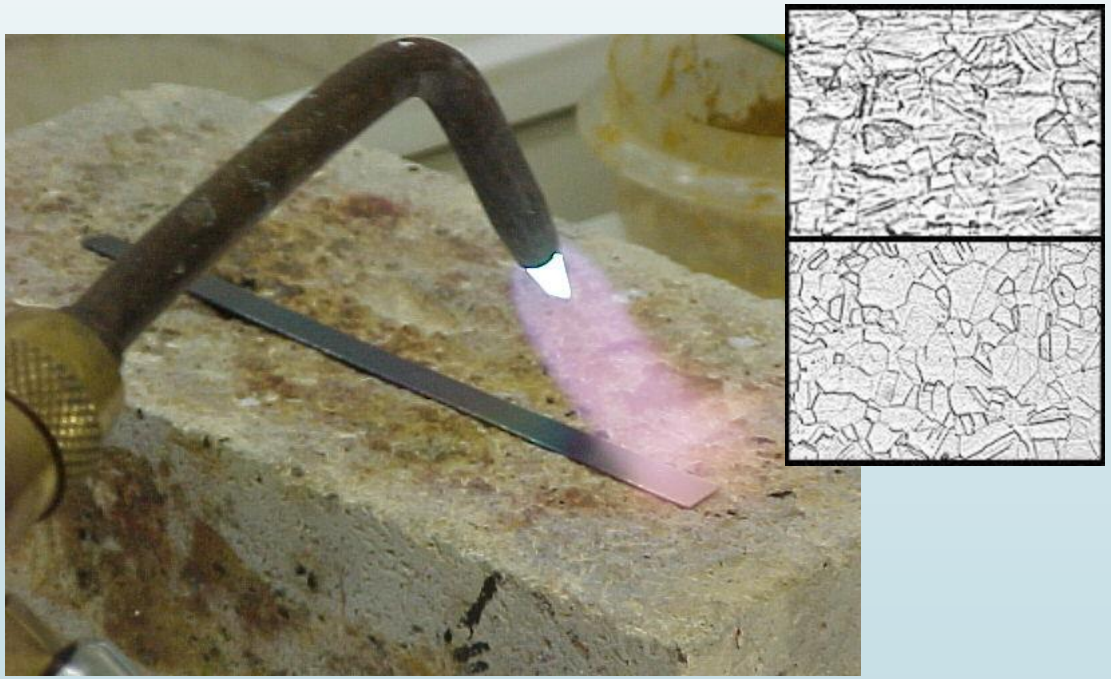
Travelling Salesman Problem – Tabu Search (Python Code)

- DEMO:

  https://smartmobilityalgorithms.github.io/book/content/TrajectoryAlgorithms/TabuSearch.html

Simulated Annealing

- Based on the properties of Metal working
- Metal is heated and cooled in order for it to be made workable



Simulated Annealing

- Start at an initial temperature value (usually something hot) and create a random initial solution
- Until temperature is cool
  - Make a small change to the current solution (neighbour)
  - Decide if we move to this neighbour solution
    - Neighbour solution is better or
    - Rand_Distribution() < temp
  - Decrease the temperature

Simulated Annealing

1. Start with a random tour through the selected cities.
2. Pick a new candidate tour at random from all neighbors of the existing tour. One way to pick a neighboring tour “is to choose two cities on the tour randomly, and then reverse the portion of the tour that lies between them.” This candidate tour might be better or worse compared to the existing tour, i.e. shorter or longer.
3. If the candidate tour is better than the existing tour, accept it as the new tour.
4. If the candidate tour is worse than the existing tour, still maybe accept it, according to some probability. The probability of accepting an inferior tour is a function of how much longer the candidate is compared to the current tour, and the temperature of the annealing process. A higher temperature makes you more likely to accept an inferior tour.
5. Go back to step 2 and repeat many times, lowering the temperature a bit at each iteration, until you get to a low temperature and arrive at your (hopefully global, possibly local) minimum. If you’re not sufficiently satisfied with the result, try the process again, perhaps with a different temperature cooling schedule.

Travelling Salesman Problem - Simulated Annealing (Pseudocode)

- **SIMULATED- ANNEALING**(*source*,*destination*, *schedule*, *num_iterations*) **return** a *route*

```
current ← random route from source to destination
for num_iterations do
T ← schedule()
neighbours ← current.NEIGHBOURS
next ← randomly choose one state from neighbours
ΔE ← next.COST - current.COST
if ΔE < 0 then current ← next
else current ← next only with probability of e^(-ΔE/T)
route ← current
return route
```

Travelling Salesman Problem - Simulated Annealing (Python Code)

- DEMO:

  https://smartmobilityalgorithms.github.io/book/content/TrajectoryAlgorithms/SimulatedAnnealing.html

  https://toddwschneider.com/posts/traveling-salesman-with-simulated-annealing-r-and-shiny/

Tabu search vs. semi-deterministic approach

- While tabu search uses a semi-deterministic approach to escape local optima, simulated annealing uses a complete stochastic approach

Visualization of metaheuristics for the TSP

- https://www.youtube.com/watch?v=rYjxtmt8g9A

Questions?